SentiFormer: Contextual Sentiment Attention for Multi-Class Social Media Sentiment Analysis

Abstract

This paper presents SentiFormer, a transformer-based model for multi-class sentiment analysis on social media text. We achieve state-of-the-art accuracy of 94.7% on the SemEval-2023 benchmark dataset. Our approach fine-tunes a pre-trained BERT-Large model with a custom attention pooling mechanism and domain-specific data augmentation. The model was implemented using PyTorch 2.1 and the Hugging Face Transformers library version 4.35. Training was performed on 4 NVIDIA A100 GPUs over 48 hours. The key contribution is our novel Contextual Sentiment Attention (CSA) layer that captures long-range emotional dependencies in text.

1. Introduction

Sentiment analysis remains a critical task in Natural Language Processing (NLP) with applications in brand monitoring, political opinion mining, and customer feedback analysis. Traditional approaches using lexicon-based methods and SVMs achieved limited accuracy of around 78-82%. The advent of transformer architectures has revolutionized this field. Our work builds upon BERT and introduces three key innovations: (1) Contextual Sentiment Attention (CSA) layer, (2) domain-adaptive pre-training on 2.3 million tweets, and (3) a multi-task learning framework that jointly predicts sentiment polarity and emotional intensity. The main objective of this paper is to demonstrate that specialized attention mechanisms can significantly improve sentiment classification accuracy beyond what general-purpose transformers achieve.

2. Related Work

Prior work in sentiment analysis can be categorized into three eras. The lexicon-based era (2005-2013) relied on sentiment dictionaries like SentiWordNet. The machine learning era (2013-2018) used SVMs and Random Forests with TF-IDF features, achieving 82-85% accuracy. The deep learning era (2018-present) introduced LSTMs and transformers. Kim et al. (2020) used BERT-Base for binary sentiment classification achieving 91.2% accuracy. Zhang et al. (2022) proposed RoBERTa-Sentiment with 93.1% accuracy. Our SentiFormer advances beyond these by introducing attention mechanisms specifically designed for emotional context understanding.

3. Methodology

Our methodology consists of four main steps. Step 1: Data Collection and Preprocessing - We collected 3.2 million tweets using the Twitter Academic API, filtered to 2.3 million after removing duplicates and non-English text. Preprocessing included URL removal, emoji conversion to text descriptions, and hashtag segmentation. Step 2: Domain-Adaptive Pre-training - We further pre-trained BERT-Large on our tweet corpus using Masked Language Modeling (MLM) with a masking probability of 15% for 3 epochs. Step 3: Contextual Sentiment Attention (CSA) Layer - We inserted a custom attention layer between BERT encoder layers 10 and 11. The CSA layer computes attention weights using sentiment-aware query-key projections: $\mathrm{CSA}(Q,K,V) = \mathrm{softmax}(Q_s * K_s^T / \sqrt{d_k}) * V$, where Q_s and K_s are sentiment-projected queries and keys. Step 4: Fine-tuning - The complete model was fine-tuned on the SemEval-2023 Task 4 dataset with AdamW optimizer, learning rate 2e-5, batch size 32, and linear warmup over 10% of training steps.

4. Dataset

We used the SemEval-2023 Task 4 benchmark dataset for evaluation. This dataset contains 58,000 annotated social media posts across 5 sentiment classes: Very Negative, Negative, Neutral, Positive, and Very Positive. The dataset was split into 42,000 training samples, 8,000 validation samples, and 8,000 test samples. Inter-annotator agreement (Cohen's kappa) was 0.87. Additionally, we used the Stanford Sentiment Treebank (SST-5) dataset containing 11,855 sentences for cross-dataset evaluation. For domain-adaptive pre-training, we used our custom corpus of 2.3 million unlabeled tweets.

5. Results

SentiFormer achieved an accuracy of 94.7% on the SemEval-2023 test set, surpassing the previous state-of-the-art by 1.6 percentage points. On SST-5, we achieved 92.3% accuracy. The F1-score (macro-averaged) was 0.943 on SemEval and 0.918 on SST-5. Ablation studies showed that removing the CSA layer reduced accuracy to 93.2% (-1.5%), removing domain-adaptive pre-training reduced it to 92.8% (-1.9%), and removing both reduced it to 91.5% (-3.2%). The model processes approximately 1,200 tweets per second during inference on a single A100 GPU. Training cost was approximately $2,400 in cloud computing resources.

6. Conclusion

We presented SentiFormer, a transformer-based sentiment analysis model that achieves 94.7% accuracy through the novel Contextual Sentiment Attention mechanism. Our results demonstrate that domain-specific attention layers combined with adaptive pre-training significantly improve sentiment classification. Future work includes extending the approach to multilingual sentiment analysis and exploring the application of CSA to other NLP tasks such as emotion detection and sarcasm identification.